

Review

89%

8

Out of 9 points

1:39

Avg Time Per Question

Your Answers:

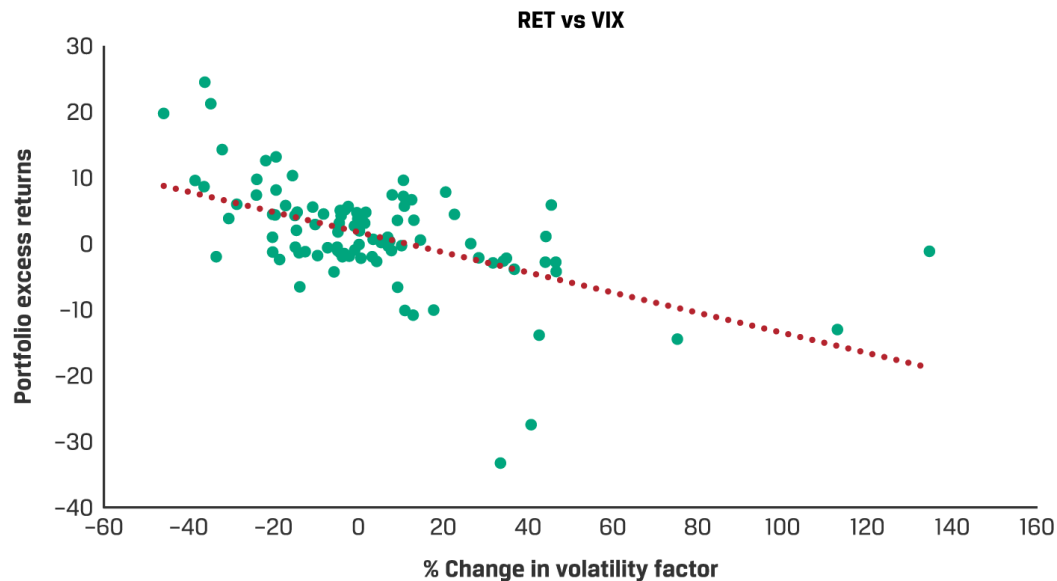
Vignette

You are a junior analyst at an asset management firm. Your supervisor asks you to analyze the return drivers for one of the firm's portfolios. She asks you to construct a regression model of the portfolio's monthly excess returns (RET) against three factors: the market excess return (MRKT), a value factor (HML), and the monthly percentage change in a volatility index (VIX).

You collect the data and run the regression, and the resulting model is

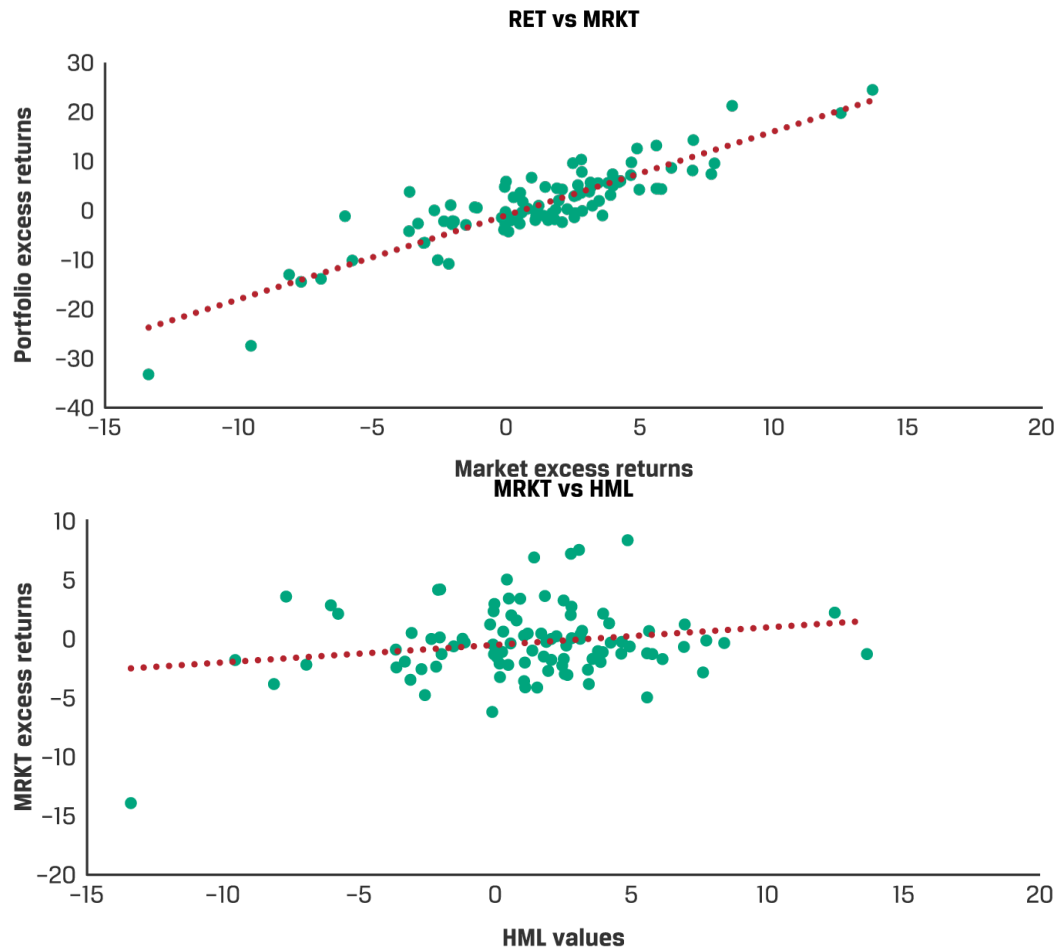
$$Y_{RET} = -0.999 + 1.817X_{MRKT} + 0.489X_{HML} + 0.037X_{VIX}$$

You then create some diagnostic charts to help determine the model fit.



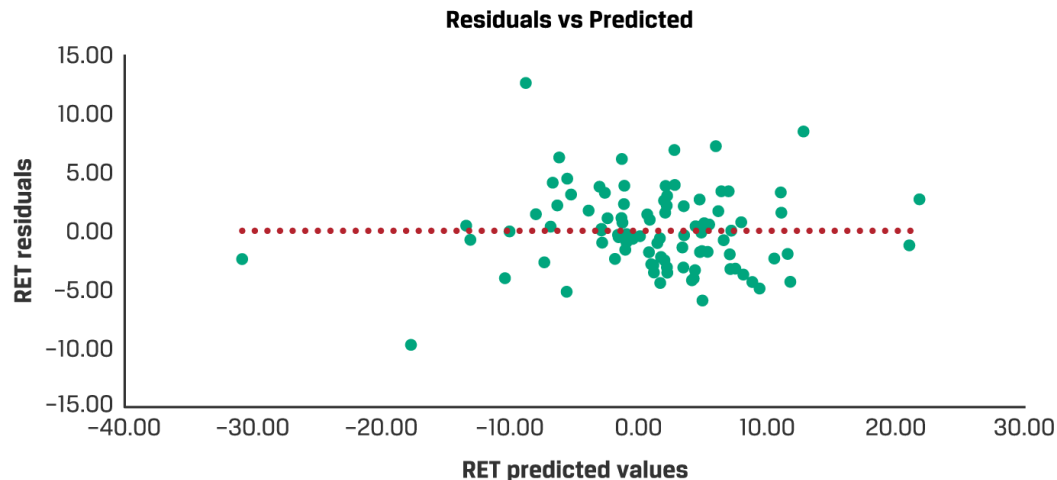
Expand / Collapse Extended Description

Percent change in volatility factor (VIX) from negative 60 to 160 on x-axis and portfolio excess returns (RET) from negative 40 to 30 on y-axis. A line falls from (negative 40, 10), (40, negative 5), to (120, negative 15). Dots surround the origin. This is a scatterplot of RET versus VIX. The relationship appears to be more curved (i.e., quadratic) than linear.



Expand / Collapse Extended Description

A scatterplot for market excess returns (MRKT) versus HML values. A flatter line intersects points (negative 5, negative 3), (0,0) and (10, 1). Dots are scattered on and around the line between 0 to 5 for HML values. This chart would most likely be used to assess the independence of these explanatory variables.



Expand / Collapse Extended Description

A scatterplot for RET predicted values on x-axis versus RET residuals on y-axis. A horizontal dotted line overlaps x-axis at (-30, 0), (0, 0), (10, 0) and (20, 0). Dots are scattered on and around the line between -5 and 10 for predicted values. This chart is very useful for visually assessing the consistency of the variance of the residuals across the observations. Any clusters of high and/or low values of the residuals may indicate a violation of the homoskedasticity assumption.

1 Multiple Choice 1 / 1 point

Determine the type of regression model you should use.

- ☐ A. Logistic regression
- ☐ B. Simple linear regression



☒ C. Multiple linear regression

Feedback

C is correct. You should use a multiple linear regression model since the dependent variable is continuous (not discrete) and there is more than one explanatory variable. If the dependent variable were discrete, then the model should be estimated as a logistic regression.

2 Multiple Choice 1 / 1 point

Determine which one of the following statements about the coefficient of the volatility factor (VIX) is true.

- ☐ A. A 1.0% increase in X_{VIX} would result in a -0.962% decrease in Y_{RET} .
- ☐ B. A 0.037% increase in X_{VIX} would result in a 1.0% increase in Y_{RET} .



☒ C. A 1.0% increase in X_{VIX} , holding all the other independent variables constant, would result in a 0.037% increase in Y_{RET} .

Feedback

C is correct. The coefficient of the volatility factor (X_{VIX}) is 0.037. It should be interpreted to mean that holding all the other independent variables constant, a 1% increase (decrease) would result in a 0.037% increase (decrease) in the monthly portfolio excess return (Y_{RET}).

3 Multiple Choice 1 / 1 point

Identify the regression assumption that may be violated based on Chart 1, RET vs. VIX.

- ☐ A. Independence of errors
- ☐ B. Independence of independent variables

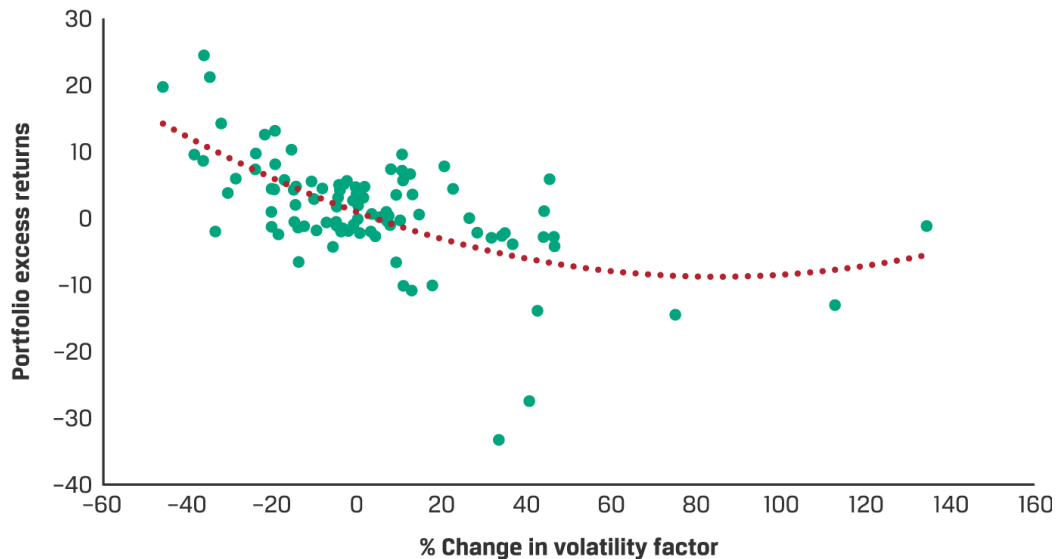


☒ C. Linearity between dependent variable and explanatory variables

Feedback

C is correct. Chart 1 is a scatterplot of RET versus VIX. Linearity between the dependent variable and the independent variables is an assumption underlying multiple linear regression. As shown in the following Revised Chart 1, the relationship appears to be more curved (i.e., quadratic) than linear.

RET vs VIX



Expand / Collapse Extended Description

% Change in volatility factor on x-axis and portfolio excess returns on y-axis. A curve falls from (negative 50, 15), (negative 20, 5) to (60, negative 5), ends at (130, negative 5). Dots are scattered on and around the curve between negative 40 to 20 for volatility factor. The relationship appears to be more curved (i.e., quadratic) than linear.

4 Multiple Choice 1 / 1 point

Identify which chart, among Charts 2, 3, and 4, is *most likely* to be used to assess homoskedasticity.

☐ A. Chart 2

☐ B. Chart 3

☒ C. Chart 4

Feedback

C is correct. To assess homoskedasticity, we must evaluate whether the variance of the regression residuals is constant for all observations. Chart 4 is a scatterplot of the regression residuals versus the predicted values, so it is very useful for visually assessing the consistency of the variance of the residuals across the observations. Any clusters of high and/or low values of the residuals may indicate a violation of the homoskedasticity assumption.

5 Multiple Choice 1 / 1 point

Identify which chart, among Charts 2, 3, and 4, is *most likely* to be used to assess independence of independent variables.

☐ A. Chart 2

☒ B. Chart 3

☐ C. Chart 4

Feedback

B is correct. Chart 3 is a scatterplot comparing the values of two of the independent variables, MRKT and HML. This chart would most likely be used to assess the independence of these explanatory variables.

Vignette

Andrew Omandi works as a senior analyst at investment firm CPSR Partners. He and junior analysts, Emmanuel Katangole, Takasongo Kasongo and Peter Mensah, have been tasked with investigating the use of multi factor models to help explain portfolio returns. After conducting some research, they identify a three-factor model described below, and are meeting to finalize their presentation to the firm's investment committee.

$$R_{it} - R_{ft} = \text{intercept}_{it} + B_M(R_{Mt} - R_{ft}) + B_{SMB}(SMB_t) + B_{HML}(HML_t) + e_{it}$$

Where:

R_{it} = portfolio return

R_{ft} = risk free rate, one-month T-bill return

B_M = market regression coefficient

R_{Mt} = return on market portfolio

$R_{Mt} - R_{ft}$ = market risk premium

B_{SMB} = SMB regression coefficient

SMB_t = return difference between small cap stocks and large cap stocks (size premium)

B_{HML} = HML regression coefficient

HML_t = return difference between high book to market stocks and low book to market stocks (value premium)

e_{it} = error term

Omandi states: "This model indicates that the main factors driving expected portfolio excess returns are premiums for market risk ($R_{Mt} - R_{ft}$), size (SMB_t) and value (HML_t). I also believe that there is a positive relationship between portfolio excess return and each of the independent variables, market risk, size and value premiums. Based on this we can formulate the following hypotheses:

Hypothesis 1

$H_o: B_M = 0$

$H_a: B_M \neq 0$

Hypothesis 2

$H_o: B_{SMB} \leq 0$

$H_a: B_{SMB} > 0$

Hypothesis 3

$H_o: B_{HML} > 0$

$H_a: B_{HML} \leq 0$

The analysts test the model and the regression results of excess portfolio returns on Mkt-Rf, SMB and HML are presented below in Exhibit 1.

Exhibit 1

Regression Statistics	
Multiple R^2	0.6235
Standard error	0.0774

Observations	60
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ANOVA	Degrees of Freedom (df)	Sum of Squares (SS)	Mean Squares (MSS)	F	Significance F
Regression	3	0.2138	0.0713	11.871	0
Residual	56	0.3362	0.0060	---	---
Total	59	0.55	---	---	---

	Coefficient	Standard Error	t-Statistic	p-value
Intercept	0.09219	0.0109	8.4577	<0.00001
Mkt-Rf	0.01348	0.00248	5.4354	<0.00001
SMB	0.0077	0.004534	1.6982	0.04829
HML	0.0043	0.00323	1.3312	0.09445

Kasango states that it is important to emphasize that the multiple linear regression model makes a number of assumptions, three of which are:

- Assumption 1: The regression residuals are normally distributed
- Assumption 2: The variance of the regression residuals is the same for all observations.
- Assumption 3: The regression residuals are correlated across observations.

Katangole asks how one can assess the goodness of fit of the estimated regression to the data.

Mensah responds, "One measure, the R^2 can be defined as the ratio of the variation in the dependent variable explained by the independent variables to the total variation of the dependent variable. However, R^2 stays the same or increases when independent variables are added to the regression. A better measure is the adjusted R^2 which does not automatically increase when independent variables are added to the regression."

6 Multiple Choice 0 / 1 point

Kasango is *least likely* correct with regard to which assumption?

☐ Assumption 1

Incorrect Answer Feedback: Incorrect because one of the assumptions of the multiple regression model is the the regression residuals are error term e_{it} is normally distributed. That is assumption 1 is correct.

☒ Assumption 2

Incorrect Answer Feedback: Incorrect because another of the multiple regression model is that the variance of the regression residuals error term e_{it} is the same for all observations. That is assumption 2 is correct.

☐ Assumption 3

Correct Answer Feedback: Correct, as there are six assumptions of the multiple linear regression model, this question addresses three of them. Two of the assumptions are stated correctly:

- Assumption 1: The regression residuals are error term is normally distributed
- Assumption 2: The variance of the residuals error term is the same for all observations.

The third one, assumption 3 is incorrect. Correctly stated it should read: The regression residuals are *uncorrelated* across observations.: the expected value of the error term, conditioned on the independent variables, is equal to zero.

Matthew Markham is a recently hired analyst at an equity research firm. Markham joins several other new employees for a training session hosted by Alexandra Garcia, a senior analyst, on the firm's approach to utilizing multiple regression in their portfolio and investment analysis.

To open the training session, Garcia explains the relationship between the dependent and independent variables in a regression model. She explains the multiple linear regression model and asks the analysts to demonstrate their understanding by confirming the assumptions for the classical model. The analysts provide the following responses about the assumptions:

- Assumption 1: The regression residuals are normally distributed.
- Assumption 2: The independent variables are not random.
- Assumption 3: The regression residuals are correlated across observations.

Garcia shares an example of a regression model completed at the company, which was based on 650 observations and 11 independent variables, and advises that a specific variable with a coefficient of 1.25 has a t -statistic of 2.39. She shares that the critical values for a two-sided t -test is 1.96 at the 0.05 significance level. Based on this information, she asks Markham to confirm if they should reject or fail to reject the null hypothesis.

In completing their review of the previous model, Garcia cautions the analysts to be aware of the various types of uncertainty that can arise when predicting the dependent variable using a linear regression model. She identifies two errors that result in the forecast of the dependent variable having a standard error larger than the standard error of the regression:

- Error1: Model error
- Error 2: Sampling error

Garcia advises that the firm prefers to use the F -test to evaluate the overall significance of a multiple regression model and asks Markham to calculate the F -statistic for a recent model that is summarized in Exhibit 1.

Exhibit 1

ANOVA	Degrees of freedom	Sum of Squares	Mean squares
Regression	2	1,924.01	962.01
Residual	122	1,683.87	13.80
Total	124	3,607.88	--
Residual standard error	--	--	0.798
Multiple R^2	--	--	0.815
Observations	--	--	125

7 Multiple Choice 1 / 1 point

Regarding the classical normal multiple linear regression model, which of the following assumptions provided by the analysts is incorrect?

☐ Assumption 1

Incorrect Answer Feedback: Incorrect because for the classical normal multiple linear regression model the required assumption is that the regression residuals are normally distributed. Expected value of the error term, conditioned on the independent variables, is 0.

☐ Assumption 2

Incorrect Answer Feedback: Incorrect because for the classical normal multiple linear regression model the required assumption is that the independent variables ($X_1, X_2, \dots, X_k$) are not random, which means that they are fixed and known.

☒ Assumption 3

Correct Answer Feedback: Correct because for the classical normal multiple linear regression model the required assumption is actually that the error term is uncorrelated across observations.

8 Multiple Choice 1 / 1 point

In the multiple regression equation $Y_i = 2.710 + 0.828 X_{1i} + 0.182 X_{2i} + \varepsilon_i$ ($i = 1, \dots, n$), for a one-unit change in X_1 , the change in Y is *closest* to:

☒ 0.828.

Correct Answer Feedback: Correct because a partial regression coefficient, b_j , describes the impact of that independent variable on the dependent variable, holding all the other independent variables constant. In this case, the partial regression coefficient for X_1 is $b_1 = 0.828$.

☐ 3.538.

Incorrect Answer Feedback: Incorrect because this is the sum of the intercept and the partial regression coefficient for X_1 , i.e. $b_0 + b_1 = 2.710 + 0.828 = 3.538$.

☐ 3.720.

Incorrect Answer Feedback: Incorrect because this is the sum of the intercept and all partial regression coefficients, i.e. $b_0 + b_1 + b_2 = 2.710 + 0.828 + 0.182 = 3.720$.

9 Multiple Choice 1 / 1 point

Which of the following is an assumption of multiple linear regression?

☒ The regression residuals are normally distributed

Correct Answer Feedback: Correct because the normality assumption requires that the regression residuals are normally distributed.

☐ The variance of the independent variables is the same for all observations

Incorrect Answer Feedback: Incorrect because the homoskedasticity assumption requires that the variance of the regression residuals (not the variance of the independent variables) is the same for all observations.

☐ There is a linear relationship between two or more of the independent variables

Incorrect Answer Feedback: Incorrect because the linearity assumption requires that the relationship between the dependent variable and the independent variables (not between two or more independent variables) is linear.

Also, the independence of independent variables assumption requires that there is no exact linear relation between two or more of the independent variables or combinations of the independent variables.